

05 · What to control for — and what NOT to (pathmc)

July 6, 2026

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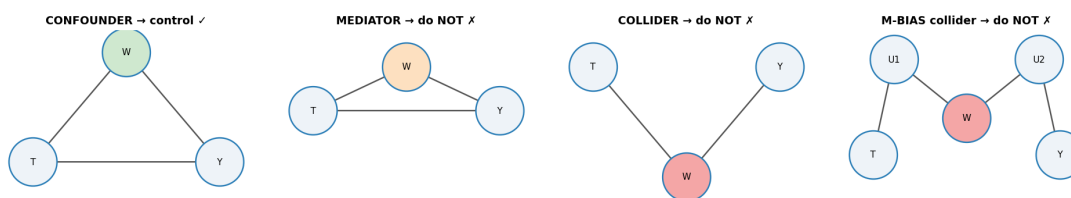
The decision. Which variables go in the model? This silently biases everything above it. Throw in “every column we have” and you’ll control for **mediators** (removing the effect you wanted) and **colliders** (inventing an effect that isn’t there); leave out a real **confounder** and you keep the bias. This is the “how did we know what to control for?” slide — the one that makes a skeptical CMO trust the rest.

We (a) build a **gallery** of the four canonical structures and the correct action for each, (b) let **pathmc** read the DAG and enumerate the admissible adjustment set + flag colliders, (c) demonstrate the bias of each wrong choice **in euros**, and (d) bound the residual risk from an *unobserved* confounder with a sensitivity contour and an **E-value**.

7-step contract, with identification (not estimation) as the whole point.

1.1 1b · The four structures — a control-or-not gallery

Every “should I control for W?” question reduces to *what role W plays* between treatment T and outcome Y:



- **Confounder** (common cause of T and Y): **control** — it opens a backdoor path.
- **Mediator** (on the causal path T→W→Y): **do not** — you’d remove part of the effect you want.
- **Collider** (common effect of T and Y): **do not** — conditioning *opens* a spurious path.
- **M-bias collider** (pre-treatment, common effect of two latents that separately cause T and Y): **do not** — even a *pre-treatment* variable can be a trap. “Adjust for everything pre-treatment” is wrong.

1.2 2 · Simulate a ground truth

email -> spend with a **true effect of €6**. Around it: **loyalty** is a **confounder** (drives both email and spend -> must control); **responded** is a **collider** (caused by both email and spend -> tempting but poisonous); **opened_email** is a post-treatment descendant.

TRUE effect of email on spend = €6.0

	email	loyalty	opened_email	responded	spend
0	1.0	1.101262	1.0	0.0	27.705363
1	0.0	0.338431	0.0	0.0	17.425501
2	0.0	-0.539972	0.0	0.0	9.602566
3	0.0	-1.260242	0.0	0.0	5.056443
4	1.0	-1.894621	0.0	0.0	7.175152

1.3 3 · Identify — let the DAG choose the adjustment set

The **backdoor criterion**: an admissible set Z blocks every backdoor path $T \rightarrow Y$ and contains no descendant of T . **pathmc** enumerates admissible sets and flags colliders straight from the graph — **no data needed**, because identification is about structure, not numbers.

Admissible adjustment set(s) for email -> spend: [{'loyalty'}]

Identifiable: True

control for {'loyalty'}: OK - no collider opened

control for {'responded', 'loyalty'}: 'responded' is a collider between 'email' and 'spend'. Conditioning on it may open a spurious path and introduce bias.

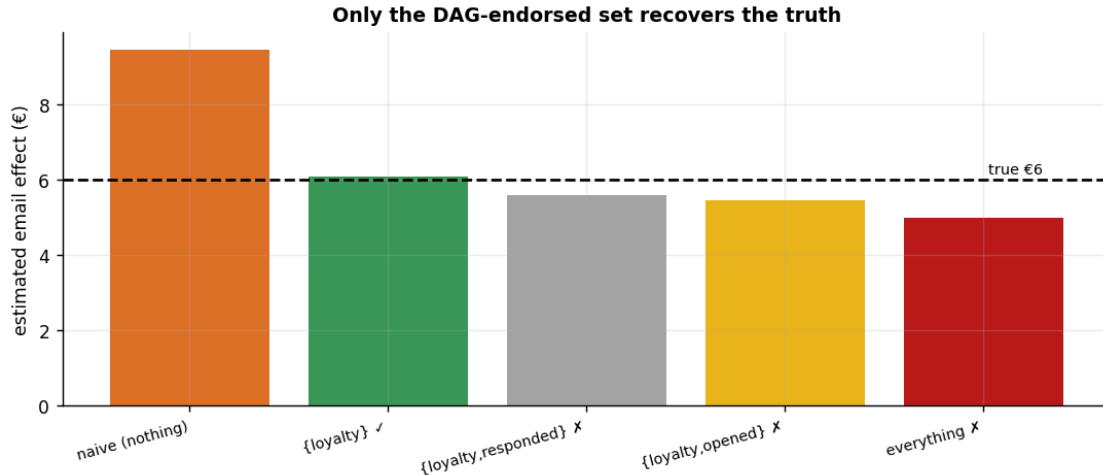
control for {'opened_email', 'loyalty'}: OK - no collider opened

control for {}: OK - no collider opened

1.4 4–5 · Estimate & Validate — every control choice, in €

We fit the email->spend effect under five control choices and compare to the true €6. Only the DAG-endorsed set ({loyalty}) recovers it; the others are biased, and **the sign of the bias is predictable from the structure** — omitting the confounder biases *up*, conditioning on a collider or post-treatment variable biases in the direction of the induced association.

naive (nothing)	€ 9.45	(error +3.45)
{loyalty} [ok]	€ 6.09	(error +0.09)
{loyalty,responded} [x]	€ 5.58	(error -0.42)
{loyalty,opened} [x]	€ 5.45	(error -0.55)
everything [x]	€ 4.97	(error -1.03)

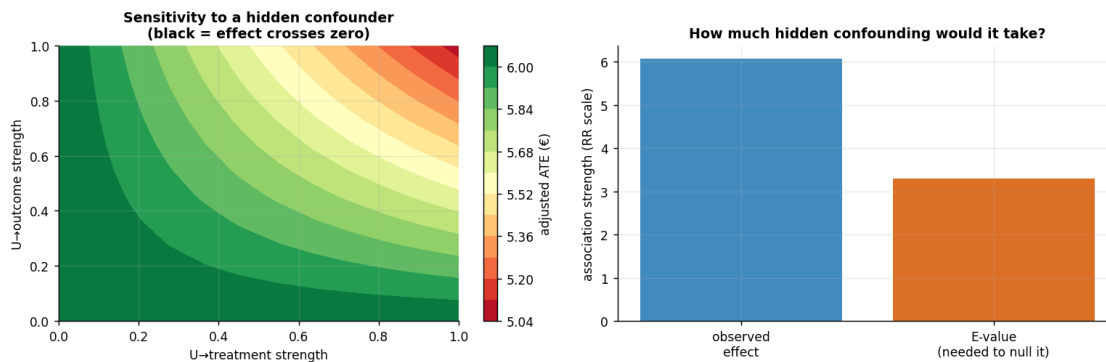


1.5 6 · Decide, in euros — with a robustness statement

The value here is a **defensible spec** we can explain to a skeptic: *here's why we controlled for loyalty, and here's why we did NOT control for opens/responses*. But identification is untestable, so we bound the residual risk: how strong would an *unobserved* confounder have to be to overturn the conclusion? `pathmc.sensitivity` gives the tipping surface; the **E-value** summarises it as a single interpretable number.

Observed adjusted ATE €6.07 (true €6.0) · E-value ~ 3.29

An unmeasured confounder would need ~3.3x association with BOTH email and spend to explain the effect away. Ask the domain expert whether anything that strong is plausible.



1.6 7 · Caveats

- **The graph is an assumption.** `pathmc` gives the correct set *for the DAG you drew*. A missing or reversed edge changes the advice — the DAG must encode defensible domain

knowledge.

- **Colliders hide in “obvious” controls.** “engaged”, “responded”, “opened”, “clicked” are almost always post-treatment; excluding them feels wrong to practitioners and is exactly right.
- **Front-door as a fallback.** If a T–Y confounder is genuinely unmeasured but a fully-mediating measured variable exists, the front-door adjustment can still identify the effect — a tool to keep in reserve.
- **Sensitivity != proof.** A high E-value means “robust to plausible hidden confounding,” not “unconfounded.” Pair it with the expert’s judgement about what could be missing.